Build and run

English version of ../../BUILD.md.

Pinned versions

Tool	Version	Source of truth
Python	3.12	`backend/Dockerfile`, CI
Backend dependencies	exact versions	`backend/requirements.lock.txt`
Node	22	`frontend/Dockerfile`, CI
Frontend dependencies	`package-lock.json`	`npm ci`
PostgreSQL	16	`docker-compose.yml`

Level A — source code

```
make doctor       # tools, .env, ports
make lint         # ruff with an explicit rule set (see ruff.toml)
make test-backend # domain and API
make test-auth    # the auth module without PostgreSQL
make docs-check
```

Level B — frontend build

```
cd frontend && npm ci && npm run build
```

Level C — full stack

```
cp .env.example .env     # POSTGRES_PASSWORD is mandatory
make compose-up
make smoke-test API_URL=http://localhost:8000
```

Level D — CI

`ci.yml` runs backend tests, the auth tests, the frontend build, the OpenAPI contract check, the documentation check and a compose configuration check. `portability.yml` adds the full-stack scenario: schema on an empty database, an idempotent restart, backup → restore → smoke test, and a start without optional integrations.

Level E — documentation as artefacts

```
make docs-pdf    # Markdown -> PDF for both languages, output in docs/pdf/
```

`tools/docs_pdf.py --check` (part of `make docs-check`) fails when a document has no translation or no PDF.